

Chapter 22. Transtracheal Gas Insufflation, Transtracheal Oxygen Therapy, Emergency Transtracheal Ventilation

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Transtracheal Gas Insufflation, Transtracheal Oxygen Therapy, Emergency Transtracheal Ventilation: Introduction

Clinical evidence highlights the importance of limiting airway pressure during mechanical ventilation. In addition, experimental results suggest the importance of avoiding lung overdistension and cyclic end-expiratory airspace collapse and reexpansion, indicating that both phenomena promote mechanical damage and release of inflammatory mediators.^{1–3} Unfortunately, interventions that can attenuate the structural insult caused by mechanical ventilation, such as the use of low tidal volume, high positive end-expiratory pressure, and reduced respiratory rate, can limit total minute ventilation.^{4,5} In this context, transtracheal oxygen therapy and tracheal gas insufflation (TGI) could have a role as adjuncts to mechanical ventilation.^{6–10}

Basic Principles

Mechanism of Action

TGI attempts to minimize dead space by delivering fresh gas through an intratracheal catheter to flush the anatomic dead space free of CO₂. During TGI, low-to-moderate flows of fresh gas introduced near the carina, either continuously or in phases, dilute the CO₂ in the anatomic dead space proximal to the catheter tip. Because CO₂ is washed out during expiration, less CO₂ is recycled back into the alveoli during the subsequent inspiration. Any catheter flow during inspiration contributes to the inspired tidal volume (V_T) but bypasses the anatomic dead space proximal (mouthward) to the catheter tip. At higher catheter flow rates, turbulence generated at the tip of the catheter by the jet stream can enhance gas mixing in regions distal to the catheter tip, thereby contributing to CO₂ removal.^{11–14} The fresh gas stream exiting the catheter tip rapidly establishes an expiratory front beyond the catheter tip between CO₂-rich alveolar gas and CO₂-free fresh catheter gas.¹³ This front is practically abolished by inverting the catheter tip and directing the catheter jet mouthward, thus eliminating the distal effect of TGI.¹² These observations indicate that the primary mechanism of CO₂ elimination during TGI is expiratory washout, and the forward-directed TGI penetrates a substantial distance into the central airways, extending the compartment susceptible to CO₂ washout with a smaller contribution of turbulence beyond the straight catheter tip. Consequently, partial pressure of arterial carbon dioxide (P_{aCO_2}) during TGI falls as a nonlinear function of catheter flow rate. Initially, modest flow rates achieve large decrements in P_{aCO_2} , but once the anatomic dead space is flushed free of CO₂, the effect on P_{aCO_2} diminishes as catheter flow rate increases.^{6,11}

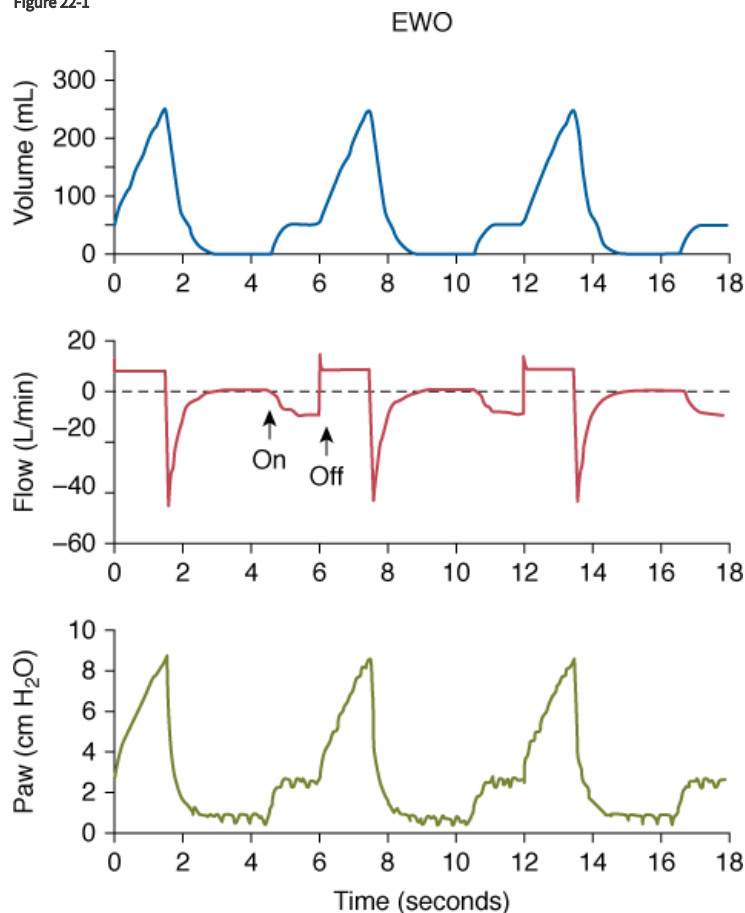
Modes of Operation

During TGI, fresh gas can be delivered continuously, or delivery can be timed to occur in phases during a specific portion of the respiratory cycle by gating a solenoid valve that either directs the flow to the catheter or diverts it to the atmosphere.^{15,16} During continuous TGI, closure of the expiratory valve during inspiration causes catheter flow to deliver variable portions of the inspired V_T .^{17,18} Phasic inspiratory TGI can be used as the only source of fresh gas, thereby bypassing the anatomic dead space proximal to the catheter tip.¹⁶ It can also be combined with a conventional ventilator to augment alveolar ventilation. During continuous or phasic inspiratory TGI, the catheter-delivered portion of the

inspired V_T is a function of catheter flow rate and inspiratory time. During phasic expiratory TGI, catheter flow is timed to occur during all or part of expiration and does not contribute appreciably to the inspired V_T .

The effect of insufflating fresh gas during specific phases of the respiratory cycle has been examined under different catheter-flow conditions. Catheter flow only during inspiration (inspiratory bypass) effectively avoids the “anatomic” dead space proximal to the catheter tip (extending from the ventilator’s Y piece). Insufflation during late expiration (expiratory washout) washes the proximal dead space free of CO_2 . Limiting TGI to the final 60% of expiration effectively reduces Pa_{CO_2} (not different from panexpiratory TGI) while limiting exposure of the trachea to TGI gas and reducing the potential for TGI-induced hyperinflation (Fig. 22-1).^{16,19,20} Although experimental studies suggest that restricting the flow of TGI gas to some portion of the expiratory phase preserves effectiveness, a rigorous engineering analysis showed that applying TGI flow solely within the final 50% of the expiratory phase yields a near maximal effect of expiratory TGI (Fig. 22-2), and this approach could simplify implementation and decrease adverse consequences.²¹

Figure 22-1

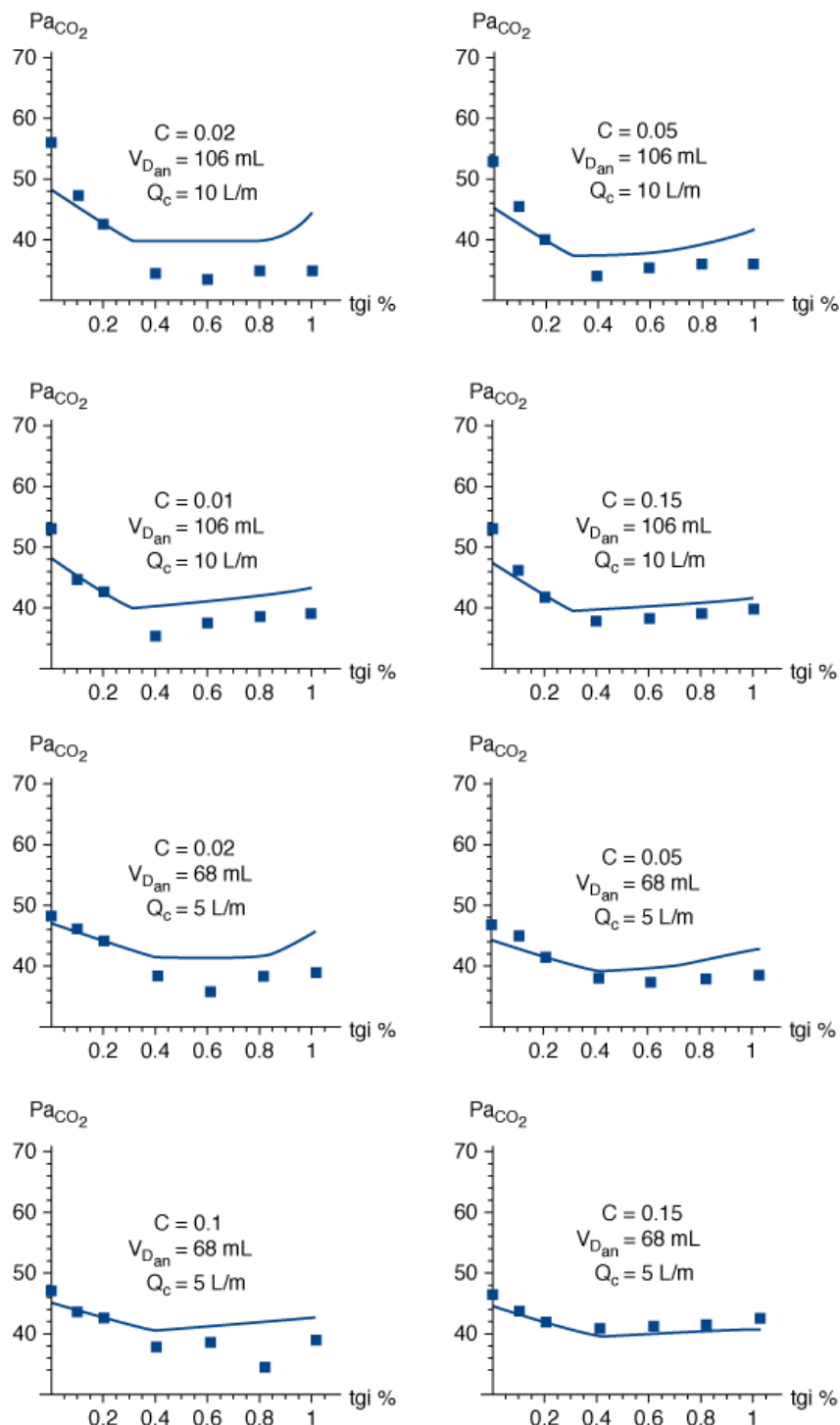


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Simultaneous tracings during expiratory washout (EWO) at 10 L/min catheter flow. (Above) Plethysmographic lung volume relative to the end-expiratory lung volume measured without catheter flow. (Center) Flow tracing measured in the inspiratory and expiratory limbs of the external circuit. (Below) Proximal airway pressure. Note that lung volume and proximal airway pressure tracings show preinspiratory step changes. These deflections indicate that gas flows both antegrade (volume tracing) and retrograde (flow tracing) from the catheter tip during this period. (Used, with permission, Burke WC, Nahum A, Ravenscraft SA, et al. Modes of tracheal gas insufflation: comparison of continuous and phase-specific gas injection in normal dogs. *Am Rev Respir Dis*. 1993;148:561–568.)

Figure 22-2



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Partial pressure of CO₂ in alveolar gas (PaCO₂) vs. duration of tracheal gas insufflation (TGI) as a percentage of expiratory time. Comparison of experimental data (boxes) with mathematical model (line). V_T = 500 mL, Re = 5 cm H₂O · L⁻¹ · sec⁻¹, and D = 0.33. V_T, tidal volume; Re, compartmental resistance during expiration. Decreasing compliance (C), elevated dead space (V_{D_{an}}), and higher catheter flow (Q_c) all magnify the effect of TGI on CO₂ clearance. Of particular note is that, with rare exception, the maximal decrease in PaCO₂ is observed at catheter flow durations between 40% and 60% of the expiratory phase, suggesting that more prolonged application of TGI accrues little additional benefit with regards to CO₂ clearance over a broad range of impedance variables. (Used, with permission, Hota S, Crooke PS, Adams AB, Hotchkiss JR. Optimal phasic tracheal gas insufflation timing: an experimental and mathematical analysis. *Crit Care Med*. 2006;34:1408–1414.)

Continuous TGI increases alveolar ventilation more than inspiratory bypass or late-expiratory washout.¹⁶ Bidirectional continuous TGI delivery produces less hyperinflation than antegrade delivery, even with small diameter endotracheal tubes (ETT).²²

Physiologic Effects

TGI reduces anatomic dead space and increases alveolar ventilation for a given frequency and V_T combination. TGI's main effect is to enhance CO_2 removal by flushing the dead space from the carina to the Y of the ventilator circuit. The catheter and the TGI jet effect, however, oppose expiratory flow and favor air trapping at end-expiration and auto-positive end-expiratory pressure (auto-PEEP).^{6,11,23–28}

TGI reduces Pa_{CO_2} during hypoventilation^{16,29–32} although TGI's efficacy in lowering Pa_{CO_2} diminishes when an increased alveolar component dominates the total physiologic dead space.^{18,32,33} An inverse correlation between respiratory rate and Pa_{CO_2} ³⁴ indicates that lower breathing frequencies (or longer expiratory times) favor TGI efficiency (reductions in Pa_{CO_2} and physiologic dead space).

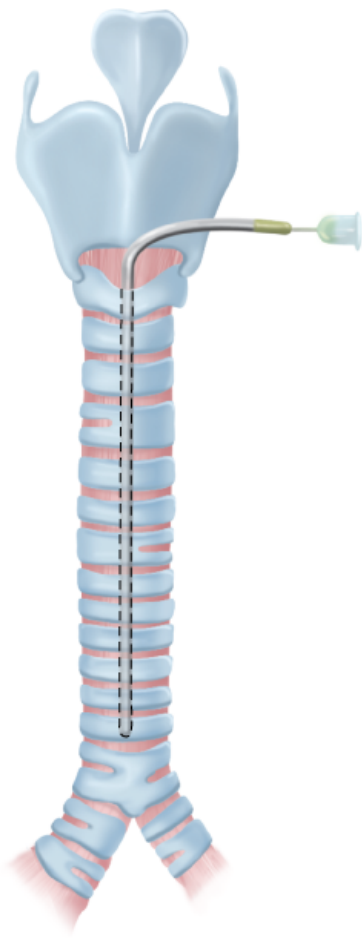
Increase in lung volume is a serious limitation of TGI and should be avoided. Solutions to minimize expiratory TGI-induced auto-PEEP include using lower TGI flows, delivering TGI during pressure-controlled ventilation (PCV), and optimizing mechanical ventilation during TGI. During PCV, a TGI-induced increase in airway pressure automatically results in decreased V_T , and the lack of expiratory TGI-induced auto-PEEP is associated with less efficient CO_2 elimination. Likewise, reducing TGI flow reduces CO_2 clearance.^{27,35–37}

Indications and Contraindications

Transtracheal Oxygen Therapy

Transtracheal oxygen therapy (TTO) administers gas directly into the trachea in nonintubated patients (Fig. 22-3). Long-term oxygen therapy is an established treatment for chronic lung disease with hypoxemia. Oxygen is usually delivered via a nasal cannula, irritating the nasal mucosa and the skin of the upper lip and ears. In 1982, Heimlich described a method for delivering oxygen directly into the trachea, bypassing the anatomic dead space and using the main airways as a reservoir.³⁸ Numerous studies have since found clear clinical benefits for TTO, including reductions in hematocrit,³⁹ pulmonary vascular resistance,⁴⁰ work of breathing,⁴¹ dyspnea,⁴² and incidence of cor pulmonale.³⁹ Other advantages are decreased oxygen cost and air pollution, decreased frequency of hospitalization, and improved exercise tolerance with better physical, psychological, and social function.⁴³

Figure 22-3



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Level of insertion and position of the indwelling tracheal catheter. (Used, with permission, Christopher KL, Schwartz MD. Transtracheal oxygen therapy. *Clin Chest Med.* 2003;24:489–510.)

The Technique

The Spofford-Christopher Oxygen Optimizing Program is a four-step protocol for TTO:⁴⁴

1. Patient evaluation, selection, and procedure preparation. Refractory hypoxemia and discomfort during maximal nasal cannula therapy are specific indications. Patients with chronic obstructive pulmonary disease (COPD) must also guarantee adequate bronchial hygiene and good pharmacologic control of airway activity throughout the program.
2. A tracheocutaneous fistula is created by a modified Seldinger technique or a surgical Lipkin procedure. Low predicted compliance with treatment (as in anxious patients) and neck anatomic alterations (as in severely obese patients) are strong contraindications to the tracheocutaneous fistula. The Lipkin method has fewer complications than modified Seldinger technique.⁴⁵
3. Tract maturation management. Patients must learn to clean the catheter and prevent inadvertent catheter displacement.
4. Mature tract management. This phase aims to prevent complications and educate the patient.

How to Ventilate the Patient

The choice of equipment for oxygen supply and delivery depends on the setting (hospital or home) and on the patient's ability to move; choices for home treatment include compressed gas cylinders, liquid oxygen, and molecular sieve oxygen concentrators. TTO usually requires an oxygen flow rate ranging from 0.25 to 1.5 L/min, but flow rates up to 2.9 L/min can be delivered through larger catheters to guarantee adequate oxygenation in severe refractory hypoxemia.⁴⁶

Oxygen flow requirements with TTO are 25-50% lower than with continuous flow therapy via a nasal cannula; 0.5 L/min of **oxygen** by the catheter is equivalent to 4 L/min by nasal prongs, because the tracheal effect acts as an anatomic reservoir that stores **oxygen** during the last part of exhalation. **Oxygen** flow requirements during exercise are also reduced by approximately 30% with concomitant reduction in patient sense of dyspnea. Moreover, the equipment is lighter and mobility may be improved.

Complications

The main complications are subcutaneous emphysema, barotrauma, and bleeding from catheter misplacement. Patients receiving long-term TTO are at risk for catheter lumen occlusion by inspissated secretions, inadvertent catheter removal (particularly dangerous during the maturation phase), and chronic tract problems like infections and keloids.⁴³

Emergency Transtracheal Ventilation

In the rare emergency situation in which it is impossible to guarantee adequate ventilation by facial mask, supraglottic device, or endotracheal intubation,⁴⁷ usually because of laryngeal stenosis, foreign bodies, tumors, or facial trauma, transtracheal ventilation is recommended.⁴⁸

The Technique

Since the first report by Jacobs et al of a series of patients who were successfully ventilated in emergency conditions using a catheter placed through the cricothyroid membrane,⁴⁹ many transtracheal emergency ventilation strategies have been reported.⁵⁰ Current techniques consist of directly inserting a 14- to 18-gauge intravenous catheter between the tracheal rings or through the cricoid membrane. The device is connected to a syringe and advanced through the skin until the airway lumen is punctured. After air aspiration confirms placement, the cannula is connected to the **oxygen** delivery system. Percutaneous transtracheal emergency ventilation is performed; dedicated kits⁵¹ or self-made devices can also be used.⁵²

How to Ventilate the Patient

Although manual or automatic jet ventilation is the reference standard for **oxygen** administration during transtracheal emergency ventilation, many techniques have been proposed to deliver **oxygen** with or without the aid of a jet valve. Commercially available resuscitation bags require great effort to ventilate through a 14-gauge catheter and make it impossible to guarantee adequate V_T .⁵⁰ Tubing that supplies **oxygen** can be coupled with a three-way stopcock, effectively creating a manual jet valve; however, tank and wall **oxygen** using flow regulators do not provide sufficient flow rates and, consequently, despite acceptable oxygenation, lead to hypercapnia. Only when the flow regulator is placed in the “wide open” position, an estimated flow of 65 L/min can be generated and physiologic V_T can be delivered.⁵⁰ Experimentally, both bidirectional manual respiration valves and active expiration using ejector-based expiratory ventilation assistance have proved useful in conditions of high expiratory resistance and occluded upper airway.^{53–55}

Complications

Complications of transtracheal emergency ventilation mainly result from the need to operate under emergency conditions and ensure ventilation as soon as possible.⁵⁶ Bleeding and posterior tracheal wall lesions are the most common, although kinking and inability to ventilate patients have also been reported when standard intravenous catheters are used.⁵⁷ Moreover, catheters equipped with safety systems sometimes do not allow a syringe to be connected. Other complications can arise with chronic therapy (e.g., infection and tracheal stenosis).⁴³

In conclusion, TTO can be an effective and safe alternative to nasal prongs for long-term home **oxygen** administration, and transtracheal ventilation using commercially available intravenous catheters and manual jet ventilation valves or wall **oxygen** supply can ensure emergency lung ventilation.

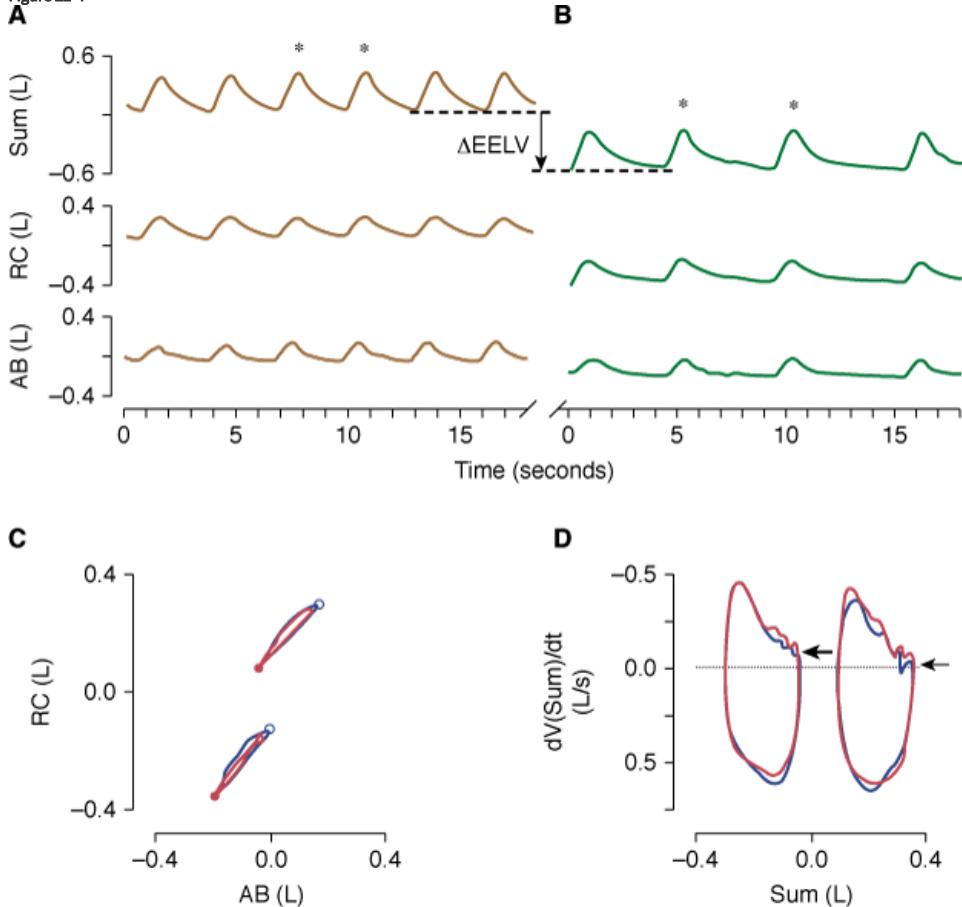
Chronic Respiratory Failure

In patients with end-stage pulmonary disease and chronic CO₂ retention, continuous insufflation of fresh gas (**oxygen** and/or air) through an intratracheal catheter has been used to provide continuous **oxygen** therapy, decrease **oxygen** flow requirements,^{7–9,58} provide a method for **oxygen** delivery, decrease dyspnea, and increase exercise tolerance.⁵⁹ A continuous low flow (4 to 5 L/min) delivered to the tracheostomy tube

reduces dead space, V_T , and minute ventilation without affecting P_{aCO_2} in the acute state and maintains or reduces P_{aCO_2} in the chronic state, presumably by reducing dead space. In patients with the most severe forms of COPD, TGI resulted in oxygen consumption and CO_2 production, as well as a less-demanding respiratory pattern.⁹

Patients with chronic respiratory failure may experience dynamic hyperinflation that may be relieved when minute ventilation decreases and expiratory time increases during high-flow insufflation. Brack et al⁶⁰ found an almost immediate decrease in end-expiratory lung volume during high-flow insufflation compared to low-flow oxygen insufflation. To investigate whether this drop in end-expiratory lung volume was caused by active expiration, they performed a Konno-Mead analysis of rib cage-abdominal volume loops. The absence of a systematic change in loop configuration and the unaffected inspiratory and expiratory asynchrony indices did not suggest increased abdominal expiratory muscle recruitment during high-flow insufflation. Therefore, the drop in end-expiratory lung volume coinciding with the reduction in minute ventilation, respiratory rate, and prolongation of expiratory time after transition to high-flow insufflation was most likely caused by the reversal of dynamic hyperinflation (Fig. 22-4).

Figure 22-4



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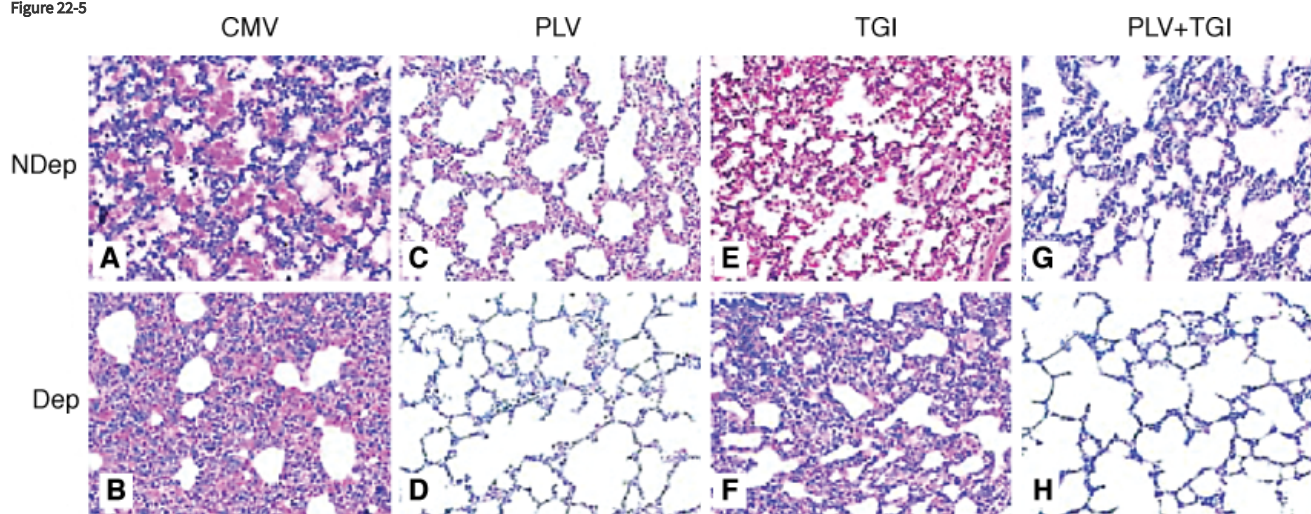
Time series of inductive plethysmographic rib cage (RC), and abdominal (AB) signals along with instantaneous lung volume (Sum) during transtracheal insufflation of oxygen at 1.5 L/min (low-flow insufflation, top left, A) and with an oxygen-air mixture at a rate of 15 L/min (high-flow insufflation; top right, B). Note the marked reduction of respiratory rate and end-expiratory lung volume (vertical line with arrow) during high-flow insufflation. For two breaths during low-flow insufflation and high-flow insufflation (marked with asterisks, respectively) rib cage versus abdominal volume loops (bottom left, C), and flow-volume loops (i.e., time derivative of the sum volume vs. the sum volume loops) (bottom right, D) are plotted. The loops at lower volumes in bottom left, C, and bottom right, D, correspond to high-flow insufflation. In bottom left, C, the major downward and minor leftward displacement of the loops with high-flow insufflation indicates that the drop in lung volume was predominantly related to deflation of the rib cage. The closed and open circles in bottom left, C, correspond to end-expiration and end-inspiration, respectively. During high-flow insufflation (top right, B), the expiratory flow approaches zero at end-expiration, as indicated by the small, horizontal arrow in bottom right, D. In contrast, during low-flow insufflation (top left, A), inspirations commence before expiratory flow has ceased (large arrow, bottom right, D), suggesting dynamic hyperinflation. (Used, with permission, Brack T, Senn O, Russi EW, Bloch KE. Transtracheal high-flow insufflation supports spontaneous respiration in chronic respiratory failure. *Chest*. 2005;127:98-104.)

Acute Lung Injury and Acute Respiratory Distress Syndrome

One of the most important features of TGI is that it can maintain normocapnia or a given level of PaCO_2 while $\dot{V}_T$ is decreased, allowing a reduction in minute ventilation. Therefore, TGI can be used to decrease the forces acting on the lung and thereby minimize ventilator-induced lung injury in patients with acute respiratory distress syndrome (ARDS).^{36,61–63}

Experimental studies on lung injury show that TGI results in lower ventilator requirements (airway pressures, $\dot{V}_T$, and dead space), more favorable alveolar surfactant composition, and a more favorable histologic trend than conventional mechanical ventilation.^{64,65} Similarly, in experimental bilateral or unilateral lung injury, the combination of TGI and partial liquid ventilation is more effective than conventional mechanical ventilation or either modality alone (Fig. 22-5).^{66–69} In this regard, TGI helps partial liquid ventilation to offset diffusional issues, remove partial pressure of arterial carbon dioxide (PaCO_2), and decrease the demands for ventilator gas pressures and $\dot{V}_T$. Perfluorocarbon liquids provide cytoprotection and low-pressure mechanical support to the atelectatic lung, and these combined effects may offset lung inflammation associated with higher $\dot{V}_T$. The combined application of TGI and partial liquid ventilation, however, is still in the early experimental stages.

Figure 22-5

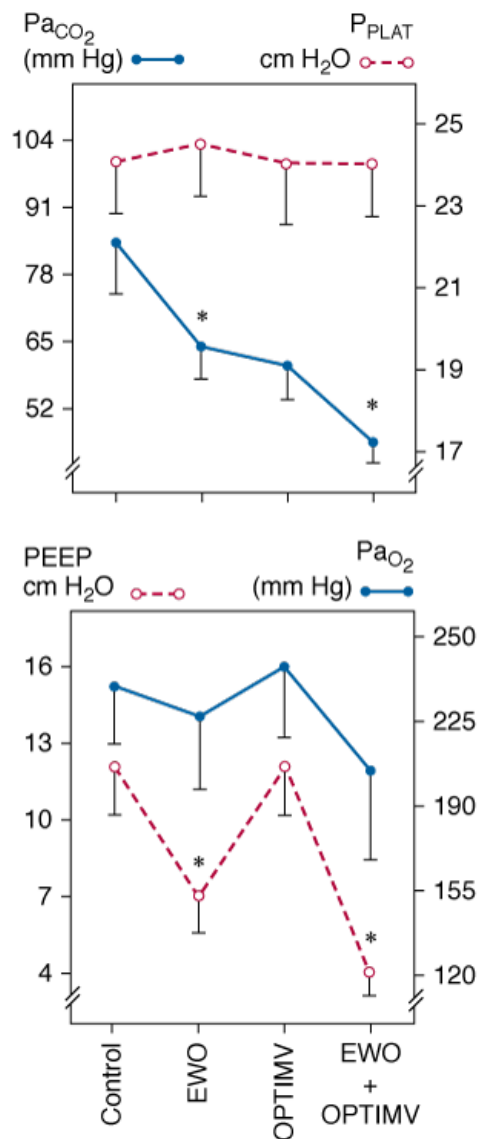


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Representative photomicrographs ($\times 100$) of lung sections obtained at 4 hours after lung injury and stained with hematoxylin and eosin. The samples were from nondependent (*NDep*) and dependent (*Dep*) regions for mechanical ventilation (*CMV*, *A* and *B*), partial liquid ventilation (*PLV*, *C* and *D*), continuous tracheal gas insufflation (*TGI*, *E* and *F*), and combination (*PLV + TGI*, *G* and *H*) groups. Intraalveolar and interstitial inflammation and hemorrhage, atelectasis, edema, and exudation were severe, especially in the dependent regions in *CMV* group. Dependent region damage was significantly less in the *PLV* and in *PLV+TGI* groups. (Used, with permission, Guo ZL, Liang YJ, Lu GP, Wang JC, Ren T, Zheng YH, Gong JY, Yu J. Tracheal gas insufflation with partial liquid ventilation to treat LPS-induced acute lung injury in juvenile piglets. *Pediatr Pulmonol.* 2010;45:700–707.)

In patients with ARDS who are ventilated with a permissive hypercapnia strategy,⁶³ the combination of increasing respiratory rate to the limit of inducing auto-PEEP, elimination of unnecessary instrumental dead space, and reduction in external PEEP when TGI-induced auto-PEEP increases (to maintain the total PEEP constant) seems a suitable pressure-limited ventilator strategy in combination with TGI^{36,70} (Fig. 22-6).

Figure 22-6



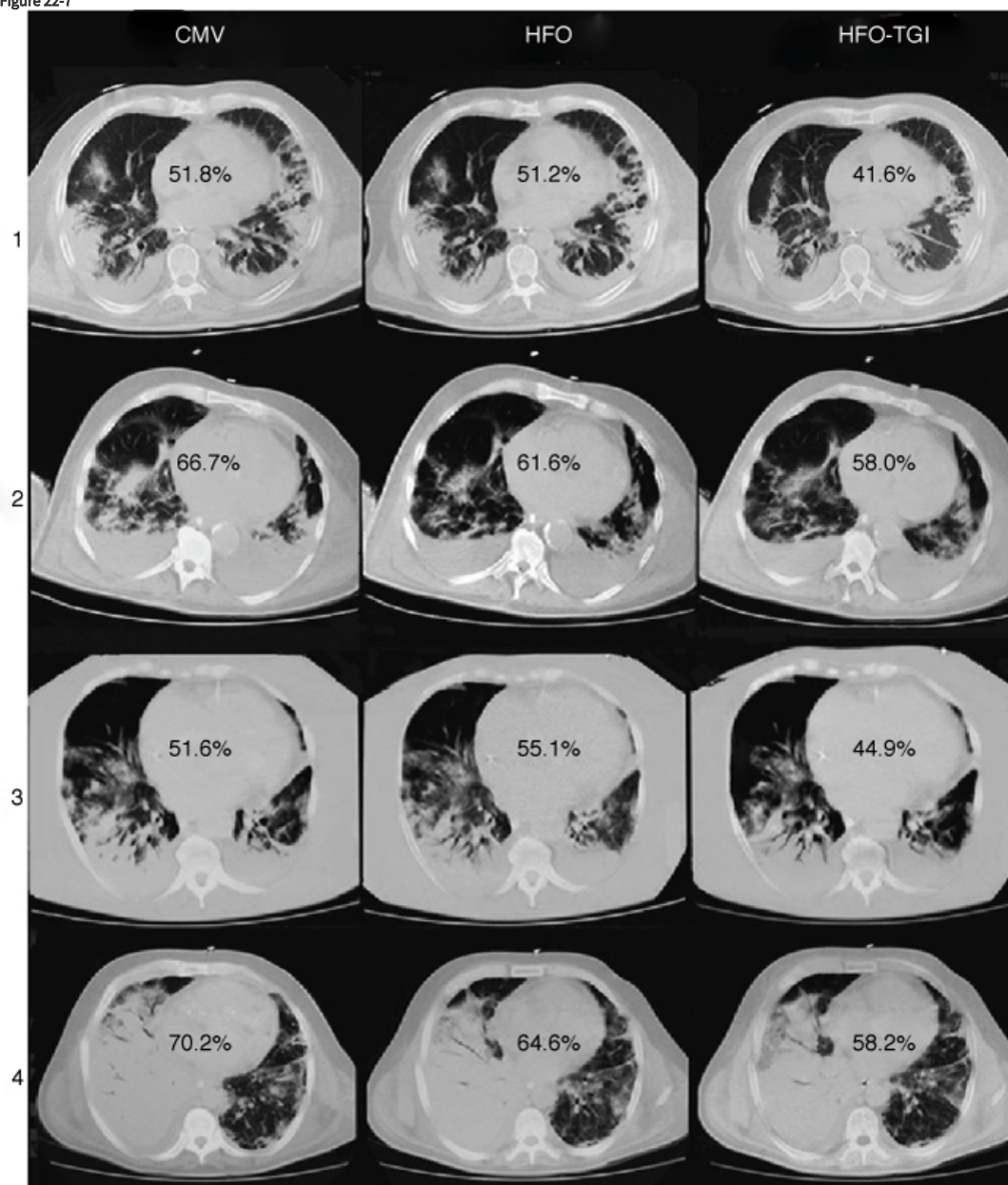
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Changes in PaCO₂, inspiratory plateau airway pressure (P_{PLAT}), PEEP, and PaO₂ induced by optimized mechanical ventilation (OPTIMV), expiratory washout (EWO), and the combination of OPTIMV and EWO in six patients with severe acute respiratory distress syndrome. Extrinsic PEEP had to be reduced by 5.3 ± 2.1 cm H₂O during EWO and by 7.3 ± 1.3 cm H₂O during the combination of OPTIMV and EWO, whereas it remained unchanged during OPTIMV alone. Plateau pressure did not change significantly, suggesting that lung hyperinflation was not produced. In patients with severe ARDS, the combination of OPTIMV and EWO has additive effects and resulted in partial pressure of arterial carbon dioxide (PaCO₂) levels close to normal values. (Used, with permission, from Richecoeur J, Lu Q, Vieira SRR, et al. Expiratory washout versus optimization of mechanical ventilation during permissive hypercapnia in patients with severe acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 1999;160:77–85.)

In ARDS, high-frequency oscillation (HFO) improves oxygenation relative to conventional mechanical ventilation. Mentzelopoulos et al recently examined whether the combination of HFO and TGI (HFO-TGI) results in better gas exchange than standard HFO and conventional mechanical ventilation.⁷¹ In fourteen patients with ARDS, the combination of HFO-TGI substantially improved oxygenation relative to both standard HFO and conventional mechanical ventilation according to the ARDS Network protocol. HFO-TGI also reduced shunt fraction and oxygenation index relative to conventional mechanical ventilation and HFO, respectively, but failed to improve PaCO₂. This study demonstrated the short-term feasibility of HFO-TGI, although its clinical utility in ventilator-induced lung injury remained uncertain. In subsequent studies, these authors compared standard HFO and HFO-TGI matched for tracheal pressure to determine whether TGI affects gas exchange independently from tracheal pressure. Compared to HFO alone, HFO-TGI resulted in a higher partial-pressure-of-arterial-oxygen-to-fractional-inspired-oxygen-

concentration ratio at similar tracheal pressures, lower PaCO_2 (at the higher tracheal pressure level),⁷² and less nonaerated lung tissue below the carina.⁷³ These results imply enhanced lung recruitment and/or gas transport and alveolar ventilation during HFO-TGI in ARDS (Fig. 22-7).

Figure 22-7



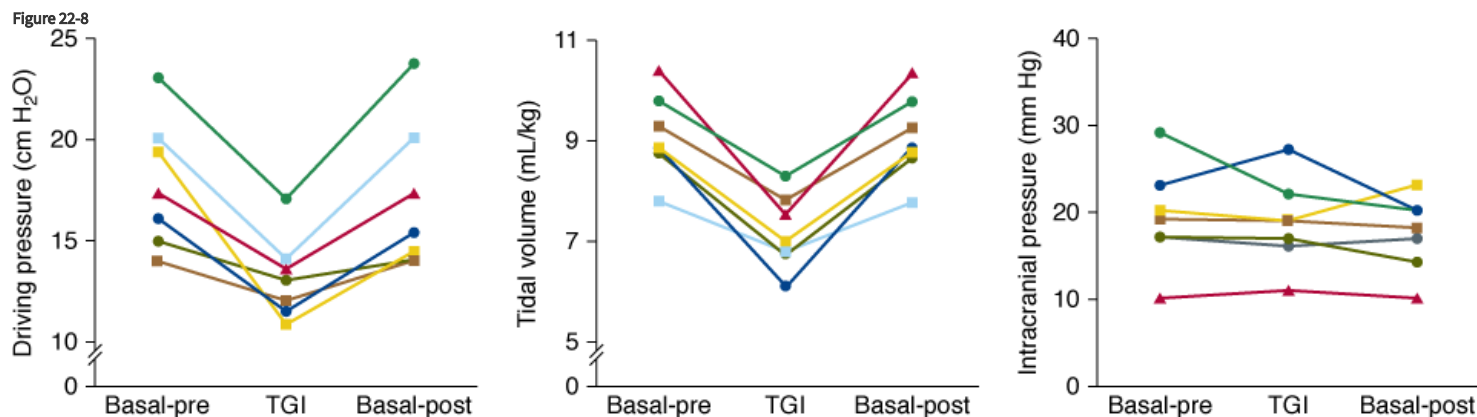
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Computed tomography (CT) sections of the lower lung from four representative patients (1 to 4). CT sections correspond to 7 cm below the carina. In patients 1 and 3, HFO preceded HFO-TGI. In patients 2 and 4, HFO-TGI preceded HFO. Percentages reflect proportions of nonaerated lung-tissue weight in lower-lung CT sections. Regional gas volume was higher during HFO-TGI versus CMV and HFO. In the nondependent lower lung, there was a HFO-TGI-related decrease in the percentage of nonaerated parenchyma versus CMV and HFO. The corresponding regional lung volume and tissue weight were higher during HFO-TGI versus CMV and HFO; HFO-TGI resulted in a higher gas volume versus CMV, and a trend toward higher gas volume versus HFO. CMV, conventional mechanical ventilation; HFO, high-frequency oscillation; TGI, tracheal gas insufflation. (Used, with permission, from Mentzelopoulos SD, Theodoridou M, Malachias S, et al. Scanographic comparison of high frequency oscillation with versus without tracheal gas insufflation in acute respiratory distress syndrome. *Intensive Care Med*. 2011;37:990-999.)

TGI-associated reduction in PaCO_2 is a potentially important maneuver in patients with cerebrovascular injury with intracranial hypertension and concomitant acute lung injury and/or ARDS, who need lung-protective ventilation and aggressive treatment to maintain intracranial pressure as low as possible. Both anecdotal case reports^{74,75} and case series⁷⁶ show that TGI in patients with acute lung injury and/or ARDS

and severe head trauma allows a more protective ventilator strategy while PaCO_2 is reduced or remains constant; more importantly, no short-term deleterious effects on hemodynamics or cerebral parameters occurs (Fig. 22-8).



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Individual values of driving airway pressure (difference between plateau pressure and PEEP) (*left*), tidal volume (*center*), and intracranial pressure before (*basal-pre*), during (TGI), and after (*basal-post*) application of expiratory TGI in patients with severe head trauma and acute lung injury. Expiratory TGI allowed the targeted PaCO_2 level to be maintained, together with substantial reductions in tidal volume and driving pressure, without deleterious effects on cerebral parameters. (Used, with permission, from Martinez M, Bernabe F, Peña R, et al. Effects of expiratory tracheal gas insufflation in patients with severe head trauma and acute lung injury. *Intensive Care Med*. 2004;30:2021–2027.)

Liberation from Mechanical Ventilation

Failure of the respiratory muscle pump is the most common cause of unsuccessful weaning from mechanical ventilation. Indeed, patients with COPD who fail weaning trials exhibit not only an almost immediate rapid and shallow breathing pattern but also progressive worsening of pulmonary mechanics with inefficient CO_2 clearance. Worsened pulmonary mechanics in these patients is characterized by increased auto-PEEP and inspiratory resistance, together with decreased dynamic lung compliance.⁷⁷ Therefore, TGI could facilitate liberation from ventilator support by enhancing CO_2 clearance.

In spontaneously breathing sheep with acute lung injury,⁷⁸ the combination of continuous positive airway pressure (CPAP) and TGI reduced the inspiratory work of breathing. The beneficial effect of TGI with CPAP on the work of breathing was attributed to a favorable balance between decreased ventilatory requirements and low workload superimposed by the apparatus and TGI. TGI may increase the work needed to open the demand valve and trigger the ventilator; this problem may be surmounted by a system that stops TGI flow before end-expiration.⁷⁹

Case series^{34,80} on the effects of TGI on lung function in patients undergoing weaning from mechanical ventilation have reported a flow-dependent reduction in $\dot{V}_T$, minute ventilation, PaCO_2 , and physiologic dead space when gas is delivered through an orotracheal tube.

Moreover, distal positioning of the TGI catheter is more effective than proximal positioning, and the effects are less pronounced in patients with tracheostomies. Interestingly, the improvement in ventilatory efficiency from the reduction of dead space yielded a decrease in PaCO_2 at the same respiratory rate and at lower $\dot{V}_T$.⁸¹

Operational Characteristics of Transtracheal Gas Insufflation

Catheter Position

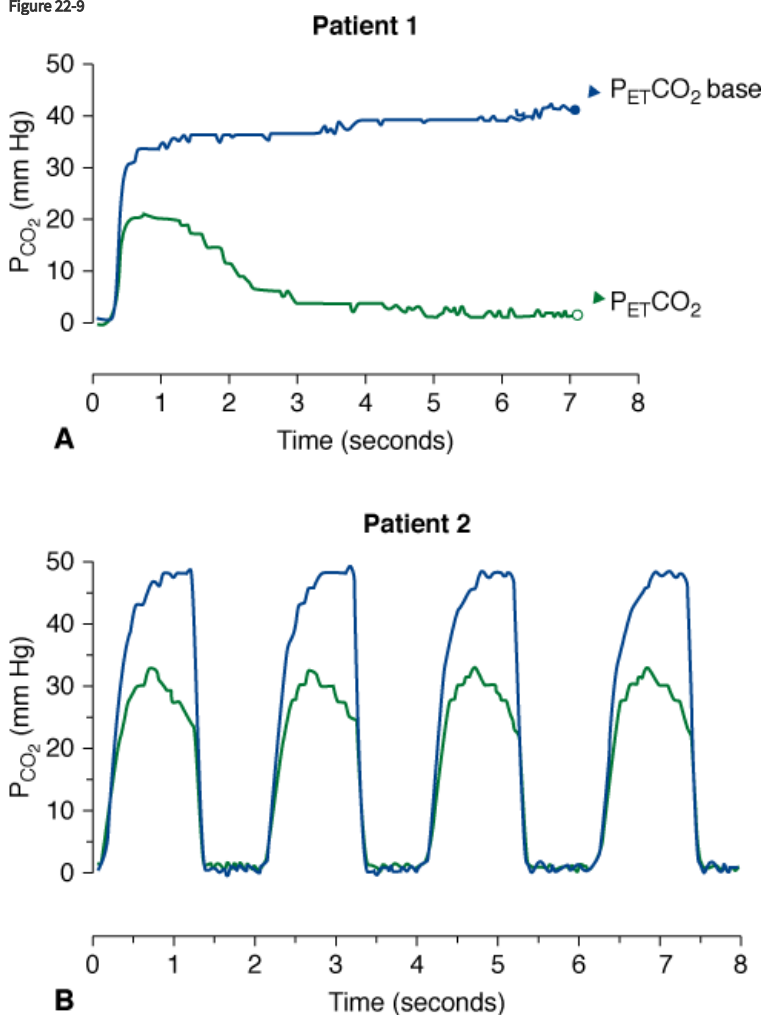
In TGI, a single catheter is usually placed above the main carina, making this technique simple to use. Nonetheless, more distal catheter placement may improve the efficiency of TGI in two ways.³¹ First, with more distal placement, a greater volume lies proximal to the catheter tip, permitting additional expiratory flushing of CO_2 -laden dead space. Moving the catheter toward the carina also advances the jet-generated turbulence zone closer to the lung periphery, thereby improving TGI efficacy. The clinical benefit, however, of introducing TGI catheters deeper than the main carina is doubtful. In a series of animal studies, the effect of TGI on PaCO_2 was strongly dependent on catheter flow rate, but

catheter tip position was not crucial provided it was within a few centimeters below or above the main carina.^{13,26,31} Similar findings have been reported in critically ill patients.⁸² Bronchoscopic guidance may not be necessary for TGI catheter placement because the position of the catheter can be verified on a chest radiograph by estimating the distance from the tip of the ETT to the main carina.

Catheter Flow Rate

TGI usually employs modest catheter flow rates. Most animal and human studies of TGI have used a flow rates of 4 to 10 L/min. CO₂ elimination during TGI depends primarily on catheter flow rate,^{31,61,82,83} with turbulence generated at higher flows enhancing distal gas mixing and CO₂ elimination.¹² Once fresh gas sweeps the proximal anatomic dead-space free of CO₂, further increases in flow rate are unlikely to wash out more CO₂. Because expiratory washout of the proximal anatomic dead space is the primary mechanism of action of TGI, both curtailing expiratory time and prolonging lung deflation may diminish the efficacy of TGI unless catheter flow rate is very high. Decreasing expiratory time would decrease the volume of fresh gas delivered to the central airways per respiratory cycle (Fig. 22-9). In situations of short expiratory times, higher flow rates are required to preserve TGI efficacy. At high catheter flow rates, the high impact pressure and shear of the inflow jet can damage the bronchial mucosa.^{84,85}

Figure 22-9



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Representative exhaled capnograms in two patients without TGI and with TGI at 6 L/min insufflation flow. A greater reduction in end-tidal CO₂ (P_{ET}CO₂) from the P_{ET}CO₂ base value corresponded with a larger reduction in PaCO₂. At a given insufflation flow, efficiency to clear CO₂ is a function of the time available to flush proximal dead space. (Used, with permission, from Ravenscraft SA, Burke WC, Nahum A, et al. Tracheal gas insufflation augments CO₂ clearance during mechanical ventilation. *Am Rev Respir Dis*. 1993;148:345–351.)

The optimal flow rate in terms of the decrement in Pa_{CO_2} afforded by TGI is a complex function of the volume of anatomic dead space proximal to the catheter tip, the volume of fresh gas delivered per expiration, the pattern of CO_2 exhalation from the lungs, and the CO_2 exchange characteristics of the respiratory system before TGI. Once dead space proximal to the catheter tip has been almost completely flushed by the fresh gas during expiration, any catheter-flow dependence of Pa_{CO_2} is likely to be secondary to enhanced turbulent mixing in the airways distal to the catheter tip. Consequently, Pa_{CO_2} continues to decrease with increasing flow rate as catheter flow rate rises but at a slower pace.^{12,31} In most TGI systems, the effect of increasing flow rate diminishes considerably when flow rate exceeds 10 L/min.

Catheter Shape

To benefit from the distal turbulence produced by TGI, the catheter must direct the jet stream toward the periphery of the lung.¹⁴ Inverting the catheter mouthward eliminates the distal effects and decreases CO_2 removal. Inverting the catheter within the ETT, however, avoids directing the jet stream onto the bronchial mucosa, which may cause bronchial injury.^{84–86} Alternatively, the catheter tip can be positioned within the ETT so that the jet hits the ETT wall. The orientation of the ETT holes with respect to the catheter (end or side) appears to have little impact on catheter efficiency.³¹ Nevertheless, catheter shape directly influences the extent (or lack of) dynamic hyperinflation caused by TGI.⁸⁷

Humidification

The fresh gas delivered by TGI should be heated and humidified to prevent mucous plug formation and to prevent TGI gas from causing bronchial injury via cooling and dehydration of the bronchial mucosa. Few studies have systematically examined the occurrence of bronchial mucosal injury during TGI. Similarly, few studies have examined the effect of conditioning TGI gas on the extent of injury to bronchial mucosa. TGI can cool tracheal gas significantly. The extent of cooling is greatest at high flow rates with continuous TGI and can be compensated only partially by conditioning the ventilator-delivered inspired gas during panexpiratory TGI.^{88,89} Case series have reported either no damage or no encrustation,⁸⁶ whereas other investigators found intratracheal catheter obstruction after 2 days of continuous use.⁶²

Endotracheal Tube Design

In most human studies, a small-caliber catheter is introduced through an angled sidearm adapter attached to the ETT and positioned just above the main carina.^{61,62,82,90} Placing a catheter through the ETT interferes with suctioning and can increase airway resistance by partially occluding the airway. Moreover, the catheter is not fixed in space and may cause injury to bronchial mucosae if it whips within the trachea at high flows. Alternatively, the catheter can be placed outside the ETT along the trachea. This technique requires visualization of the vocal cords and deflation of the ETT cuff and risks puncturing the cuff.

Designs incorporating channels within the ETT wall would solve these problems and simplify TGI application. Boussignac et al^{91,92} embedded small capillaries in the walls of an ETT for TGI, and also used this modified ETT to deliver high-velocity jets of O_2 at the carinal orifice to prevent arterial O_2 desaturation during suctioning.⁹³ The modified ETT can be used to make mechanical ventilation less aggressive in different clinical scenarios.^{94,95} Future clinical applications of TGI will probably use a modified ETT that incorporates the catheter in its wall attached to a standardized circuit for gas delivery. In any case, TGI should never require reintubation.⁹⁶

Other systems allow aspiration of anatomic and instrumental dead space in the late part of expiration, and replacing the aspirated volume with fresh gas through the inspiratory line of the ventilator improves CO_2 clearance.⁹⁷ This aspiration system has allowed reductions in airway pressure and V_T while keeping Pa_{CO_2} constant in healthy humans,⁹⁸ as well as in patients with ARDS⁹⁹ and COPD.¹⁰⁰ This aspiration system might avoid the problems associated with jet streams of gas or with gas humidification without developing auto-PEEP.

Adjustments at the Bedside

Inspired Oxygen Fraction

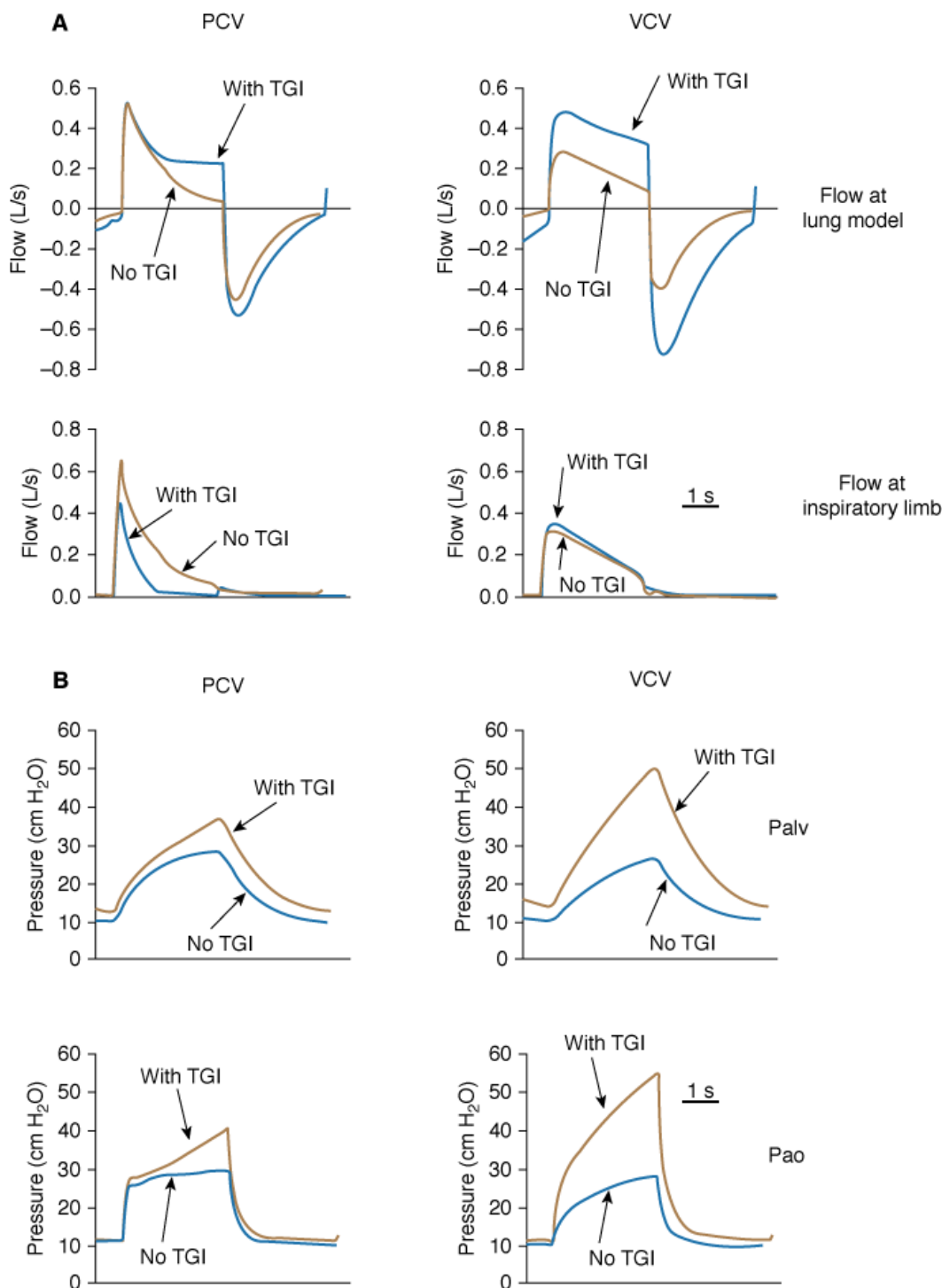
The actual fraction of inspired carbon dioxide (FI_{O_2}) during TGI depends on two factors: the contribution of TGI to total inspired V_T and the FI_{O_2} of the FI_{O_2} catheter gas. If, however, the of the catheter gas is matched to ventilator FI_{O_2} , the actual inspired FI_{O_2} will always be identical to

that delivered by the ventilator.

Airway Opening Pressure

During TGI, the jet stream increases flow through the ventilator circuit during expiration and creates a region where bidirectional flows exist. Both effects change the resistance characteristics of the respiratory system and modify the relationship between airway opening (P_{ao}) and alveolar (P_{alv}) pressures observed at baseline (catheter flow rate, 0 L/min). Because expiratory resistance increases during TGI, P_{ao} tends to underestimate P_{alv} (and functional residual capacity [FRC]) when the system is switched from baseline to TGI conditions. During panexpiratory TGI, catheter flow ceases during inspiration, and inspiratory P_{ao} provides as much useful information regarding P_{alv} as during conventional mechanical ventilation. In contrast, during continuous TGI, inspiratory P_{ao} (measured at the tip of the ETT) differs from the tracheal pressure (Fig. 22-10). During expiration under both continuous and expiratory TGI conditions, however, catheter flow pressurizes the respiratory system, so P_{ao} underestimates tracheal pressure.⁸⁴ During expiration, the extent that P_{ao} underestimates tracheal pressure increases with catheter flow rate and depends on the geometry of the system and the orientation of the catheter with respect to the trachea.

Figure 22-10



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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A. Flow-versus-time tracings of delivered gas flow measured both at airway opening and distal to the entrance of tracheal gas insufflation (TGI) during both pressure-controlled ventilation (PCV) and volume-controlled ventilation (VCV) with and without the addition of 12 L/min TGI in a lung model. **B.** Pressure-versus-time tracings of system pressure measured both at the airway opening (*Pao*) and distal to the entrance of the TGI flow (*Palv*) during both PCV and VCV with and without the addition of 12 L/min of TGI flow in a lung model. Regardless of whether VCV or PCV was used, continuous TGI increases *Pao* and peak carinal pressure (*B*). In association with these changes there is an increase in tidal volume. During VCV (*A*), the flow from TGI is additive to the flow from the ventilator. During PCV, the addition of the TGI flow caused the flow from the ventilator to decelerate more rapidly; once ventilator flow reached zero, a square wave flow pattern derived entirely from the TGI system persisted. (Used, with permission, from Kacmarek RM. Complications of tracheal gas insufflation. *Respir Care*. 2001;46:167–176.)

Effect of Transtracheal Gas Insufflation on Lung Volume

Catheter flow delivered during inspiration contributes to total inspired V_T . This contribution is eliminated if TGI is timed to occur only during expiration.¹⁶ In most TGI circuits, however, this volume is small (approximately 10 to 20 mL at a catheter flow rate of 10 L/min). The effect of TGI on total inspired V_T depends on the ventilator mode and on FRC in a flow-dependent fashion.^{12,82} TGI can increase FRC in three ways. First, part of the momentum of the discharging jet stream is transferred to the alveoli.¹⁰¹ Second, the catheter decreases the cross-sectional area of the trachea, increases expiratory resistance, and delays emptying. Third, catheter flow through the ETT, expiratory circuit, and expiratory valve can build up a backpressure that impedes deflation and is the major determinant of dynamic hyperinflation.¹⁰² Continuous TGI increases FRC more than expiratory TGI, especially when the inspiratory time fraction is prolonged.¹⁰²

Dynamic hyperinflation caused by TGI may represent either a problem or a therapeutic option³⁶ and can be manipulated by using an inverted-jet insufflator to achieve a venturi effect.^{12,86} Monitoring dynamic hyperinflation during TGI requires a means of external lung-volume measurement, such as impedance plethysmography. Guided by the plethysmograph signal, ventilator-set PEEP can be adjusted to maintain FRC constant as flow rate is varied.^{26,32} Alternatively, if the TGI system allows an end-expiratory hold maneuver, the ventilator-set PEEP can be adjusted to maintain total PEEP constant.^{37,103} Alternatively, during PCV, a flow-relief valve automatically compensates for the extra gas introduced into the system by TGI and eliminates the need for ventilator adjustments to control total PEEP.¹⁰⁴

Transtracheal Gas Insufflation–Ventilator Interactions

During flow-controlled, volume-cycled ventilation, total inspired V_T can be maintained relatively constant during continuous TGI by decreasing the ventilator-set V_T .^{26,61} During PCV, TGI application does not change the total inspired V_T , provided TGI does not pressurize the respiratory system beyond the set pressure. As catheter flow rate increases, ventilator-delivered V_T declines, but the total inspired V_T remains the same.^{26,83} If inspired V_T delivered by the catheter (V_{Tc}) exceeds the V_T generated by PCV in the absence of TGI, then TGI will overpressurize the circuit, and peak P_{ao} will be greater than that produced by the ventilator-set pressure. Consequently, excessive pressures can be produced within the respiratory system if V_{Tc} is too large. When this happens, the P_{ao} -time profile becomes a hybrid of PCV and constant-flow volume-cycled ventilation, resembling that generated during volume-assured pressure-support ventilation. This problem can be circumvented by introducing a pressure-release valve into the ventilator circuit that dumps circuit pressure above a set threshold.¹⁰⁵

The interactions between TGI and ventilator mode, V_T , and P_{ao} that result in development of auto-PEEP deserve mention. During volume-controlled ventilation, when ventilator PEEP is left constant, V_T remains constant, but end-expiratory pressure and P_{ao} increase because of TGI-induced auto-PEEP. During PCV, when ventilator PEEP and peak airway pressure are kept the same as baseline, V_T excursions (and hence minute ventilation) are reduced because of TGI-induced auto-PEEP. When ventilator PEEP is reduced by an amount equivalent to TGI-induced auto-PEEP, V_T , peak airway pressure, and total PEEP remain the same as baseline during PCV^{37–88}.

During CPAP delivered by a mechanical ventilator in combination with TGI, additional inspiratory effort is required to overcome the insufflation flow and trigger the ventilator valves. Bench studies have found that TGI might interfere with ventilator triggering at low peak inspiratory flow rates, suggesting that weak patients may fail to open the demand valve at high catheter flow rates.¹⁰⁶

Monitoring

Careful monitoring of delivered volumes and pressures is necessary to ensure safe application and to evaluate the effect of TGI on lung function. A catheter inside the ETT may increase both inspiratory and expiratory resistances, particularly when small endotracheal or tracheostomy tubes are used.^{6,11,32,107,108}

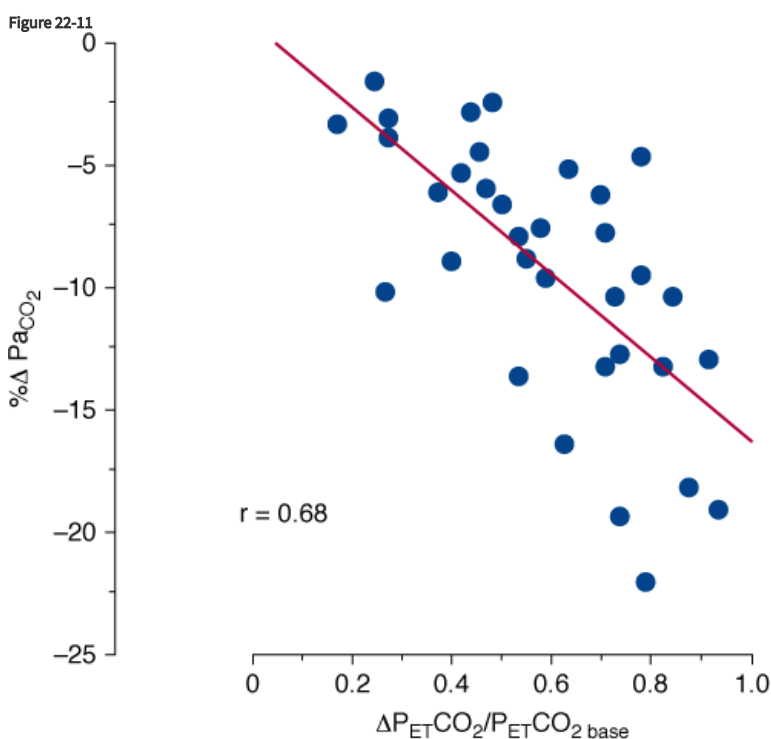
Because TGI introduces an external flow source independent of the ventilator, it can hinder the ventilator's ability to monitor pressures and volumes and may cause the ventilator alarm to go off incessantly. Catheter flow during expiration disables the monitoring role of the ventilator's expiratory pneumotachograph, triggering ventilator alarms when the difference between the measured inspired and exhaled volumes exceeds a certain value. More importantly, external flow that can pressurize the ventilator circuit interferes with the ventilator's ability to detect leaks. Using the end-expiratory occlusion technique to measure auto-PEEP may increase lung volume dramatically if TGI flow is not

interrupted simultaneously. The same effect will occur with continuous TGI during end-inspiratory occlusions.^{6,30} TGI may also interfere with clinicians' ability to measure lung mechanics like respiratory system compliance and auto-PEEP.

Continuous-flow TGI could increase delivered V_T , airway and alveolar pressures, and total PEEP with both PCV and with volume-controlled ventilation. Moreover, in PCV, when ventilator flow reaches zero, continuous flow from continuous TGI increases V_T and airway pressures.

These increases occur because the exhalation valve of the ventilator is not active during the inspiratory phase.²⁴ Overpressurization can be identified by examining the airway pressure tracing,¹¹ and can be remedied by placing a pressure-relief valve in the ventilator circuit to dissipate insufflated flow that produces excess pressure.^{78,109} Complete obstruction of the outflow can cause overinflation of the lungs in seconds, with the potential for pneumothorax or hemodynamic compromise.

The efficacy of TGI can be monitored by capnography. Expiratory capnograms provide an indicator of the effect of TGI on the CO_2 concentration in the gas remaining in the proximal anatomic dead-space compartment at the onset of inspiration.^{62,82,90} Although end-tidal P_{CO_2} is a poor estimate of P_{aCO_2} ^{110,111} in patients with respiratory failure, changes in end-tidal P_{CO_2} induced by TGI correlate significantly with changes in and justify routine measurement of end-tidal P_{CO_2} during TGI^{33,62,82,90} (Fig. 22-11).



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Percentage reduction in arterial P_{CO_2} (P_{aCO_2}) from baseline as a function of the reduction in partial pressure of end-tidal P_{CO_2} (P_{ETCO_2}) from the baseline value (P_{ETCO_2} base). As the difference between P_{ETCO_2} and P_{ETCO_2} base widened, larger reductions in arterial P_{CO_2} were observed. (Used, with permission, from Ravenscraft SA, Burke WC, Nahum A, et al. Tracheal gas insufflation augments CO_2 clearance during mechanical ventilation. *Am Rev Respir Dis*. 1993;148:345–351.)

Unknowns

The delivery of catheter gas at higher flows must be examined with regard to the need for humidification and the potential for tracheal damage with long-term use. Only inconclusive, limited data are available on the clinical safety of TGI.^{85,87,112} Turbulent gas conditions promote shear stress, increased gas impact on the airway walls, and the transfer of a higher kinetic energy to the tracheal mucosa.^{101,103} The end-hole TGI catheter mode could theoretically cause more airway damage than large-caliber reverse-thrust catheters because the flow exiting the end-hole

catheter is closer to the carina and points directly at it. Further studies are needed to assess clinical safety before broad clinical application.^{113–114}

The Future

TGI remains a promising technique that is only used as an investigational or late option at the bedside. TGI should be weighed against established invasive and noninvasive modes of ventilation. Evidence demonstrating better patient–ventilator interaction and the absence of significant adverse effects must be accumulated before TGI can be considered suitable for standard intensive care practice. Future TGI devices must include certain features:^{30,84,105,109,115} (a) good coordination with the ventilator, (b) monitoring capabilities like automatic TGI flow shutoff to prevent overpressurization of the ventilator circuit and airways, (c) contextual alarms for TGI settings and TGI-ventilator–patient interaction, and (d) commercially available devices and catheters.

Summary and Conclusion

TTO therapy is safe for long-term administration of home oxygen. Transtracheal ventilation, using commercially available intravenous catheters and manual jet ventilation valve or wall oxygen supply, can ensure emergency lung ventilation. Experimentally, TGI is very effective at reducing lung volume while maintaining similar levels of PaCO₂ during permissive hypercapnia in patients with ARDS. Similarly, TGI has been successfully used in patients with concomitant severe brain and lung injury and in patients with chronic respiratory failure. Nevertheless, concerns regarding patient application, safety, monitoring, and interaction with the ventilator remain. Further investigations are necessary before TGI can be routinely employed in intensive care units.

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